

APIx: Evaluator Evidence & Differentiation

Dossier

Real-time Airfare Price Index for India | Problem Statement: SIH 26056

MoSPI / RBI / DGCA

149/149 Tests Passing

1. EXECUTIVE VERDICT & THE FOUR PROOF PILLARS (ALL INDIA-SOURCED)

Verdict: YES — Statistically Defensible & Grounded in Official Indian Records. Foreign comparisons have been eliminated. The entire problem and solution are anchored directly in Parliamentary admissions, MoSPI's official CPI Base 2024=100 documentation, DGCA statistics, and Gazette aircraft rules.

 **Parliamentary Admission**

DGCA TMU covers only **78 routes (~27%)** on random monthly spot-checks. Minister confirmed on Rajya Sabha floor that monitoring capacity must be strengthened [E5, E5b].

 **Enforcement Void**

Dec 2025 Price Caps: ₹18k statutory cap breached at **₹64,783** (DEL-TRV) in Parliament Zero Hour while CPI recorded 0.09% inflation [E6, E6b, E7].

 **MoSPI Policy Precedent**

CPI Base 2024=100: MoSPI created 12 online web markets and launched a Data Innovation Lab hiring web scraping roles for official statistics [E1, E2].

 **Academic/OTA Distinction**

Cleartrip built deal advice; SkyPredict built ML prediction. APIx builds the missing national index with DGCA weights & quarantine [E9, E10].

2. THE INDIA-ONLY DIFFERENTIATION MATRIX

Capability	MoSPI CPI 2024	DGCA TMU Unit	MoCA Price Caps	Cleartrip Trends	SkyPredict (ML)	APIx (Our System)
Primary Purpose	Inflation Deflator [E1]	Tariff Compliance [E5]	Crisis Control [E6]	Consumer Deals [E9]	ML Research [E10]	Official-Grade Index
Automated Engine	Not stated (online) [E1]	✗ Manual checks [E5]	✗ Not documented	✓ Live search query	✓ Batch Python [E10]	✓ Scheduled Playwright
Frequency	Weekly web / Monthly [E2]	Monthly random [E5]	Ad-hoc in crisis [E6]	At user search	One-off batch [E10]	✓ Daily continuous cron
Booking Windows	Not published [E1]	Not stated [E5]	✗ Uncontrolled	Up to 60 days [E9]	Generic feature [E10]	✓ Fixed T+7 & T+30
Route Coverage	Not published [E1]	78 routes (27%) [E5]	Crisis routes only	User searched	6 metro cities [E10]	✓ DGCA-ranked corridors
Traffic Weights	Not published [E1]	✗ No index [E5]	✗ None	✗ None	✗ None	✓ DGCA Pax Weights [E11]
Fee Unbundling	Not published	✗ None	✗ None	✗ None	✗ None	✓ Base + Tax + UDF Split
Published Output	Group 07.3 Index [E3]	None public [E5]	Ministerial orders	On-screen badge	Journal Paper [E10]	✓ Daily Series + Dashboard
API Access	eSankhyiki portal [E8]	✗ None	✗ None	✗ None	✗ None	✓ 11 REST Endpoints

APIx: Evidence Dossier & Evaluation Defense

Evidence Citations, Past Solution Flaws & Presentation Script

RFC 9309 Compliant

Zero Base Drift

3. OFFICIAL EVIDENCE CITATIONS (ALL INDIAN SOURCES)

[E1, E2] MoSPI CPI Base 2024=100 (PIB 12 Feb 2026) FAQ Q7, Q14, Q26, Q27: Airfares collected from online platforms. 12 online web markets created. Sample sizes and routes not published.	[E3, E4] MoSPI Transport Index 07.3 & eSankhyiki Airfares lumped with rail, bus, and auto (Index 103.50, +2.17% in Jan 2026). eSankhyiki portal provides macro data but no standalone airfare basket.
[E5, E5b] Rajya Sabha Admissions (8 & 15 Dec 2025) MoS Murlidhar Mohol: DGCA TMU checks 78 routes (~27% traffic) manually on monthly basis. Minister Naidu: TMU requires strengthening.	[E6, E6b, E7] Dec 2025 MoCA Fare Caps & Disruption ₹18k cap breached at ₹64,783 (DEL-TRV) in Parliament. MMT quotes reached ₹82,000 (DEL-MAA) while CPI reported only 0.09% transport inflation.
[E9, E10] Cleartrip (2026) & SkyPredict (IOSR 2024) Cleartrip launched "Price Trends" deal advice. SkyPredict scraped 65,000 rows across 6 cities for ML regression—neither built a price index.	[E11] DGCA FY23–24 Domestic Passenger Traffic Official city-pair statistics: DEL-BOM (43.8%), DEL-BLR (32.2%), BOM-BLR (24.0%). Used as locked, auditable weighting constants in APIx.

4. WHY PREVIOUS SOLUTIONS FAILED VS. HOW APIX WINS

Failure Mode in Past Solutions	What Previous Teams Built & Why Juries Rejected It	APIx Architectural Advantage
1. The "Flight Deal Alert" Trap	Built Telegram/Email price drop bots. Rejected because B2C shopping tools do not serve MoSPI/RBI macroeconomic inflation needs.	National Macro Deflator: Computes an axiomatic price index with policy summary barometers and elasticity analytics.
2. The "Black-Box ML" Trap	Trained XGBoost/LSTM to predict future fares. Rejected because ML predictions cannot be audited by NSO or defended under UN/ILO CPI rules.	Axiomatic Index Engine: Jevons geometric mean with monthly chain-linking, satisfying Time-Reversal & Transitivity tests.
3. The "OTA Scraper" Trap	Scraped MakeMyTrip/Ixigo. Rejected for violating OTA robots.txt and including dynamic convenience fees (₹350–₹500).	RFC 9309 RobotsGate: Ingests directly from Air India & IndiGo public search endpoints without OTA markup distortions.
4. The "Naive Average" Trap	Simple arithmetic mean $\frac{\sum P_i}{N}$. Rejected due to extreme substitution bias and equal weighting of small regional routes.	DGCA Traffic Weights: Weighted by official city-pair passenger volumes across fixed T+7 and T+30 advance windows.
5. The "Fake Data" Trap	Hardcoded JSON dummy curves. Rejected when evaluators asked to see live ingestion or database rows.	Database Quarantine & Health Telemetry: Synthetic gap-fillers quarantined in view; live "Trigger Ingestion" writes real DB rows.

 **30-Second Winning Pitch to Deliver to the Evaluator:**

"Previous attempts built consumer flight-deal bots, black-box ML predictors, or naive scrapers that violate robots.txt. **APIx is national economic infrastructure**—combining RFC 9309 compliant multi-source ingestion, DGCA traffic-weighted geometric index math, strict synthetic data quarantine, and an auditable 8-page dashboard designed specifically for MoSPI and RBI."